

ANIMAL PULSE SENSING

User Guide

Flexible Animal Pulse-Sensing Hardware

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1. Introduction

The Animal Pulse Sensing board is a flexible, battery-powered embedded sensing platform designed to acquire pulse-related sensor data and motion information using a compact microcontroller-based architecture. The hardware integrates the CH591D microcontroller, MAX86916 sensing device, BMI270 inertial sensor, multiple regulated supply rails, a USB data/debug interface, and basic status indication.

This guide is written for a non-specialist technical user. It focuses on how the board is powered, connected, programmed, and operated, rather than providing a component-by-component circuit design review.

The hardware description is based on the supplied project schematic. Firmware-dependent behavior and information not established by the schematic are identified as such rather than being presented as fixed hardware requirements.

2. System Overview

The board operates from a single rechargeable battery input in the 3.6 V to 4.2 V range. Power is controlled by the ON/OFF section and distributed to the regulated supply sections. The CH591D is the main controller and communicates with the sensing devices. A logic shifter provides voltage-domain translation where required.

- CH591D main microcontroller and system controller.
- MAX86916 pulse/optical sensing subsystem; this guide covers its supply and I²C interface only.
- BMI270 inertial sensing subsystem; this guide covers its supply and digital interface only.
- 3 V regulator regulated 3 V supply for the main digital domain.
- 1.8 V LDO regulated low-voltage rail.
- 5 V step-up converter generates the 5 V rail shown in the schematic.
- Logic shifter translates digital signals between applicable voltage domains.
- 32 MHz crystal external clock reference shown for the CH591D.
- Battery and ON/OFF circuitry portable power source and system power control.
- Debugging pads expose 5 V, GND, D+ and D– for programming/debugging.

3. Power Architecture

The board is designed around a single battery input. The required battery voltage range is 3.6 V to 4.2 V. The battery connects to the battery section and is controlled by the ON/OFF power switch before being used by the rest of the circuit.

- Battery input: 3.6–4.2 V.
- 3 V regulator: provides the regulated 3 V supply for the main digital system and associated circuitry.
- 1.8 V LDO: provides the regulated 1.8 V rail for low-voltage circuitry.
- 5 V step-up: generates the 5 V rail shown in the schematic.
- Logic shifter: translates digital signals between the applicable logic-voltage domains.
- Switch the board OFF before connecting or disconnecting the battery.

Recommended power-up sequence:

1. Confirm the battery is within the 3.6–4.2 V requirement.
2. Confirm the ON/OFF switch is OFF.
3. Connect the battery with the correct polarity.
4. Turn the ON/OFF switch ON.
5. Allow the regulated rails to settle before beginning normal sensor operation or programming.

Power-down:

6. Stop active data acquisition or programming.
7. Turn the ON/OFF switch OFF.
8. Disconnect the battery if the board will be stored or transported.

4. Main Hardware Interfaces

4.1 CH591D Microcontroller

The CH591D is the main controller of the board. It interfaces with the sensing devices, controls the board's status indication, and provides the USB data/programming connection exposed on the debugging pads. A 32 MHz crystal is shown as the external clock source.

4.2 MAX86916 Interface

The MAX86916 section is connected to the board's regulated supply and digital communication network. The schematic shows local supply decoupling and I²C pull-up resistors. For normal operation, ensure the required supply rails are present and the I²C pull-ups are connected to the appropriate logic supply.

4.3 BMI270 Interface

The BMI270 is connected to the CH591D through the digital sensor interface shown in the schematic and has local supply decoupling. Sensor communication should remain within the logic-voltage domain established by the board.

4.4 Logic Shifter

The logic shifter translates digital signals between the voltage domains used by the connected devices. Its two supply sides are connected to the corresponding rails shown on the schematic.

4.5 Crystal / Clock

A 32 MHz crystal is connected to the CH591D clock pins with associated capacitors to ground. Avoid unnecessary probing or loading of these clock nodes during normal operation.

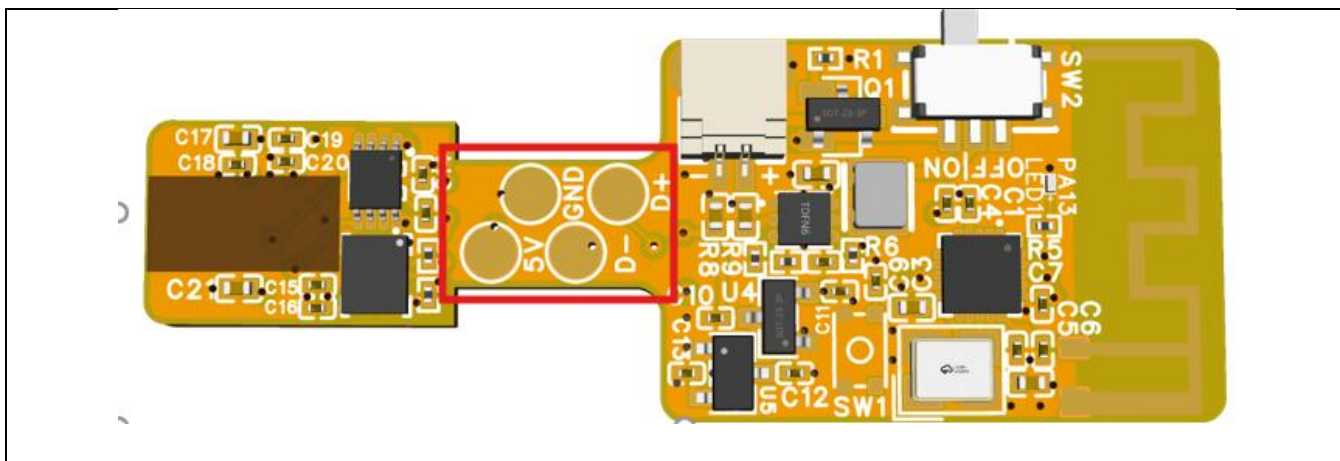
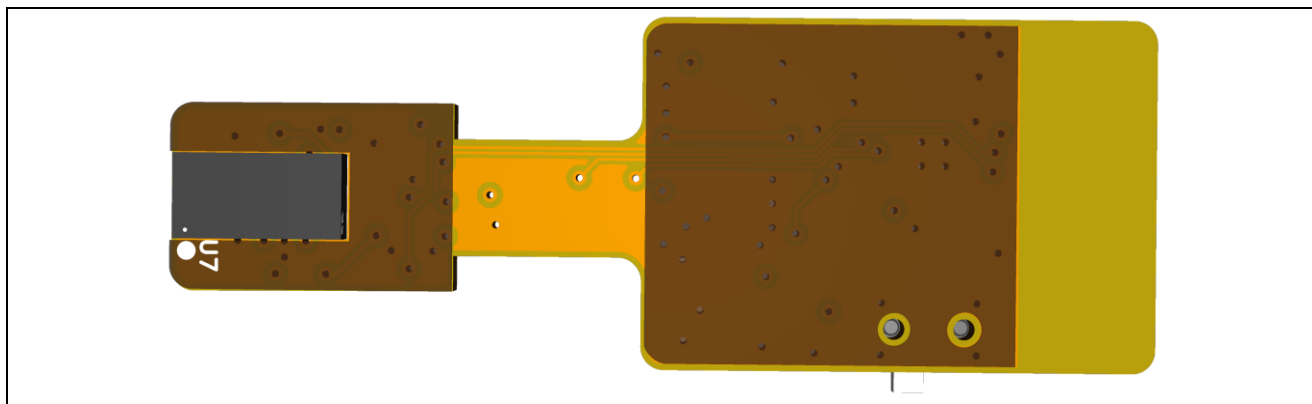
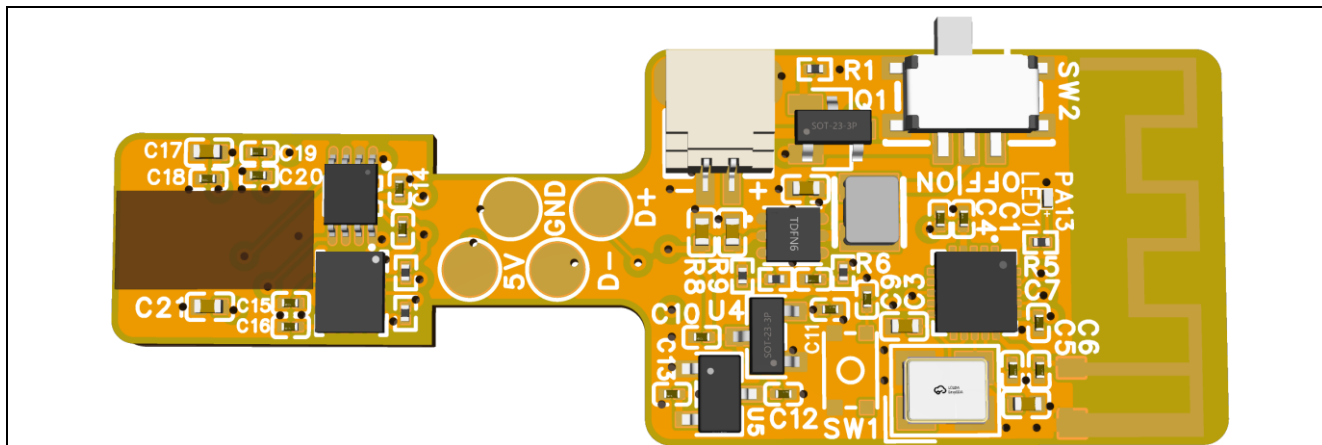
6. Signal / Pin Description

The following is a practical user-facing signal summary. It is not intended to replace the full CH591D pinout.

Signal / Net	Function	Typical Connection / Use	Notes
VBAT / BAT+	Battery input	3.6–4.2 V battery	Observe polarity.
SW_PWR	Switched power	Output of ON/OFF power path	Present when board is ON.
3 V	Regulated supply	Main digital domain	Verify before sensor operation.
1.8 V	Regulated supply	Low-voltage domain	Used by applicable circuitry.
5 V	Boosted supply	5 V domain / programming reference	Do not short to other rails.
SDA	I ² C data	MCU ↔ sensor devices	Logic shifter already added
SCL	I ² C clock	MCU ↔ sensor devices	Logic shifter already added.
D+	USB data +	Programming/debugging	Connect to USB host D+.
D–	USB data –	Programming/debugging	Connect to USB host D–.
GND	Ground reference	Battery / USB / measurement reference	Common circuit reference.
ANT	RF/antenna connection	CH591D RF section	Keep as designed.

7. Board Layout Reference

The final client version should include photographs or CAD views of the flexible board so the user can identify the battery connection, ON/OFF switch, programming pads, sensor area, and other external connection points.



8. Operating Procedure

Use the following procedure for normal operation:

1. Inspect the board for visible damage, contamination, or loose connections.

2. Confirm that the battery voltage is between 3.6 V and 4.2 V.
3. Make sure the ON/OFF switch is OFF.
4. Connect the battery with correct polarity.
5. Turn the ON/OFF switch ON.
6. Verify that the board powers up normally. If required, the 3 V, 1.8 V and 5 V rails can be checked with a multimeter.
7. Allow the CH591D firmware to initialize the sensor interfaces.
8. Operate the board according to the application's data-acquisition procedure.
9. When finished, stop application activity and switch the board OFF.
10. Sampling rate, data format, LED behavior, and application-level processing are firmware-dependent and are not fixed by the supplied schematic.

9. Programming and Debugging Using D+ / D–

The schematic provides a dedicated debugging pad group labelled 5 V, D+, D– and GND. These pads expose the USB data connection of the CH591D and provide a convenient connection point to a USB host for programming/debugging.

9.1 Pad Identification

Pad	Purpose
5 V	USB/programming power reference or board 5 V rail, depending on the programming setup.
GND	Ground/reference connection to the USB host or programmer.
D+	USB positive data line.
D–	USB negative data line.

9.2 Programming Connection

- Disconnect and remove the battery from the board.
- Switch the board OFF before making programming connections.
- Connect the USB host/programmer ground to the board GND pad.
- Connect USB D+ to the board D+ pad.
- Connect USB D– to the board D– pad.
- Connect programming power as required by the chosen programming method. If USB 5 V is used to power the board, do not simultaneously connect the battery unless the power-path design explicitly permits simultaneous sources.
- Place the CH591D into the programming/bootloader mode required by the installed firmware and programming environment.
- Start the appropriate CH591D programming tool on the host computer and select the detected USB device, if applicable.
- Program the firmware and verify that the operation completes successfully.
- After programming, disconnect the programming setup or leave it connected for debugging as required.
- Return the board to normal battery-powered operation.

9.3 Debugging

For debugging, use D+, D– and GND and keep the board supply within the intended power range. If the USB connection is not detected, first check D+/D– orientation, GND continuity, board power, and whether the installed firmware/bootloader supports the intended USB programming method.

Note: the supplied schematic identifies the physical USB data pads but does not specify the exact host-side programming software, bootloader entry sequence, or firmware tool settings. These items should be finalized from the CH591D programming documentation and the project's firmware procedure.

10. LED Indication

The schematic includes an LED connected to an MCU-controlled signal. It can therefore be used by firmware as a status indicator. The exact meaning of steady, flashing, or off states should be defined by the firmware documentation.

11. Safety and Handling

Use only a battery within the specified 3.6–4.2 V range and observe correct polarity. Do not short the battery or power rails, and avoid connecting or disconnecting programming power while another power source is active unless the power-path arrangement has been verified. Protect the flexible board from excessive bending, ESD, moisture, and mechanical damage.